

Landing-Zone Probability Model

Use repeated test data to estimate the probability that a payload lands inside a target zone.

Advanced**2 class periods****Aerospace Mission**

Engineering Context

A relief drone drops small emergency packages. The team must determine whether the current deployment method is reliable enough for a defined landing zone.

What You Will Practice

- Use descriptive statistics
- Model target success probability
- Visualize two-dimensional test data
- Communicate risk and limitations

What You Need

- Approved drop or launch simulation
- Target grid
- At least 20 trial coordinates
- Spreadsheet
- Engineering notebook

Student Instructions

- 1 Define the target-zone boundaries and coordinate origin.
- 2 Collect or analyze at least 20 landing coordinates.
- 3 Plot every point on a scaled scatter plot.
- 4 Calculate mean x and y position and spread in each direction.
- 5 Count successful landings and calculate observed success rate.
- 6 Identify directional bias and possible environmental causes.
- 7 Recommend an aim-point adjustment or process change.
- 8 Predict the expected number of successes in 50 future drops.

Engineering Requirements

- At least 20 trials
- Target success defined before data collection
- All points shown, including misses
- Include observed probability and directional bias
- Recommendation must acknowledge sample-size limits

Constraints & Safety

- Use a safe simulation or approved low-energy system
- Maintain a clear test zone
- Do not alter data after seeing target results

What You Submit

- Coordinate data
- Scatter plot
- Success-rate calculation
- Bias analysis
- Aim-point recommendation
- Future-performance prediction

Reflection Questions

- 1 Did the mean landing point fall inside the target?
- 2 Can a high average accuracy still produce poor reliability?
- 3 How would wind change the distribution?
- 4 What sample size would increase confidence?

Engineering Tip

Keep both coordinate directions; reducing the result to distance alone can hide a directional bias.

Common Mistake

Assuming the observed percentage is an exact future probability ignores limited sample size.

Stretch Goal

Create an elliptical landing zone that contains approximately 80% of the observed points.

Career Connection

Guidance, navigation, and test engineers quantify dispersion and probability for payload delivery, landing, and release systems.